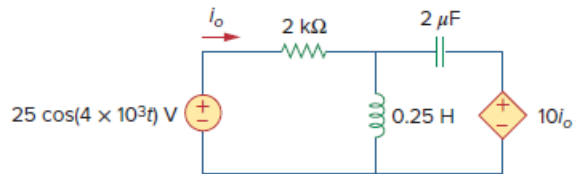


Problem

Find i_o in the circuit of Fig. 10.54.

Figure 10.54:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.54 in the text book.

The value of source voltage is,

$$v(t) = 25 \cos(4 \times 10^3 t) \text{ V}$$

From the source voltage, the angular frequency is,

$$\omega = (4 \times 10^3) \text{ rad/s}$$

The phasor representation of source voltage is,

$$V = 25 \angle 0^\circ \text{ V}$$

Determine the impedance of the $2 \mu\text{F}$ capacitor.

$$\begin{aligned} Z_c &= \frac{1}{j\omega C} \\ &= \frac{1}{j(4 \times 10^3)(2 \mu\text{F})} \\ &= -j125 \Omega \end{aligned}$$

Determine the impedance of the 0.25 H inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(4 \times 10^3)(0.25) \\ &= j1000 \Omega \end{aligned}$$

Step 2 of 4

Draw the phasor circuit for the circuit shown in Figure 10.54.

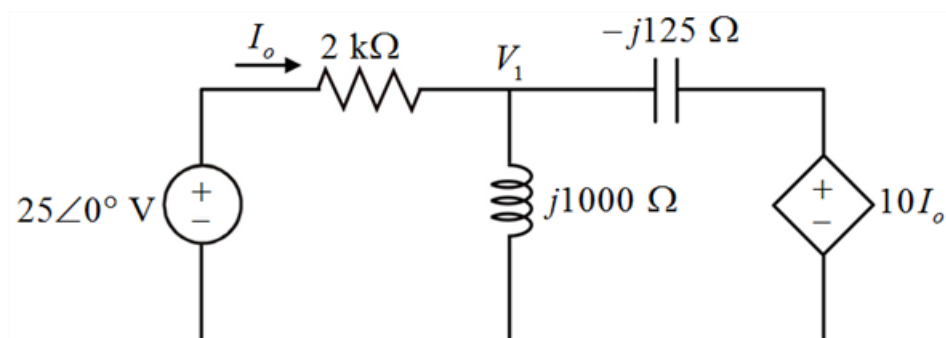


Figure 1

Step 3 of 4

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1}{j1000} + \frac{V_1 - 25\angle 0^\circ}{2000} + \frac{V_1 - 10I_o}{-j125} = 0$$

$$\frac{V_1}{j1000} + \frac{V_1 - 25\angle 0^\circ}{2000} + \frac{V_1 - 10\left(\frac{25\angle 0^\circ - V_1}{2000}\right)}{-j125} = 0$$

$$\frac{V_1}{j1000} + \frac{V_1}{2000} + \frac{V_1}{-j125} + \frac{10V_1}{(-j125)(2000)} = \frac{25\angle 0^\circ}{2000} + \frac{250}{(-j125)(2000)}$$

Further simplifications as follows,

$$V_1 \left(\frac{1}{j1000} + \frac{1}{2000} + \frac{1}{-j125} + \frac{1}{(-j125)(200)} \right) = \frac{25}{2000} + \frac{250}{(-j125)(2000)}$$

$$V_1 \left[(5 \times 10^{-4}) + j(7.04 \times 10^{-3}) \right] = 0.0125 + j0.001$$

$$V_1 = \frac{0.0125 + j0.001}{[(5 \times 10^{-4}) + j(7.04 \times 10^{-3})]}$$

$$V_1 = (0.2668 - j1.7566) \text{ V}$$

Step 4 of 4

From Figure 1, the value of current I_o is,

$$I_o = \frac{25\angle 0^\circ - V_1}{2000}$$

Substitute $(0.2668 - j1.7566) \text{ V}$ for V_1 in the equation.

$$I_o = \frac{25\angle 0^\circ - (0.2668 - j1.7566)}{2000}$$

$$= \frac{24.7332 + j1.7566}{2000}$$

$$= 12.398 \times 10^{-3} \angle 4.06^\circ \text{ A}$$

$$= 12.398 \angle 4.06^\circ \text{ mA}$$

In Figure 10.54, the voltage source is in the form of cosine.

Therefore, the time-domain representation of current is,

$$i_o(t) = 12.398 \cos(4 \times 10^3 t + 4.06^\circ) \text{ mA}$$

Thus, the time domain expression for current $i_o(t)$ is, $12.398 \cos(4 \times 10^3 t + 4.06^\circ) \text{ mA}$.